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(請填入甄選證號)

臺北市立第一女子高級中學

112 學年度科學班甄選入學

語文能力檢定試題本

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讀完本頁的說明，聽從監試委員的指示才開始作答！

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這是北一女中科學班甄選入學語文能力檢定試題本，內容包含國文與英文，提供答案卷 2 份，請分科作答。測驗時間從 **10：10** 到 **11：50**，共 **100** 分鐘。作答開始與結束請聽從監試委員的指示。

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方格內填寫甄選證號，再翻頁作答

國文試題

請閱讀以下文字，按照題意要求，完成一篇作文。

大約十年前一個夏天的晚上，我在紐約街頭散步，很多已經打烊了的商店，櫥窗兀自光亮地招展著，在一個比較陰暗的角落，有面橫幅的廣告特別顯眼：「Imagination is more important than knowledge.——A. Einstein」（想像力比知識重要——愛因斯坦）。

說這句話的人是愛因斯坦，他為什麼會這麼說？我查過他說這句話的上下文，大意是：愛因斯坦說他相信直覺與靈感的力量。他之所以說「想像力比知識重要」，是因為知識是受局限的，而想像力則可以擁抱整個世界，刺激進步，產生演化。所以愛因斯坦說：想像力是科學研究裡的一個真實元素。

但我更相信「知識比想像力重要」，是有另一個根據。牛頓說：「如果說我看得比別人更遠，那是因為我站在巨人的肩膀上。」巨人的肩膀，是需要借助知識的梯子才能站得上去的。如果一個人整理不出自己的知識系統與架構，怎麼可能站到巨人的肩膀上？所以我想，人生苦短，若非得選擇一個重點不可，那寧可堅持「知識比想像力重要」。

（改寫自郝明義〈想像力、知識和飛行〉）

在科學的發想與創造的過程中，知識與想像力缺一不可。然而，依據上文所言，愛因斯坦與牛頓對二者的重視程度不同，你比較認同哪一方的說法？請就以下兩個立場：

（一）「想像力比知識重要」

（二）「知識比想像力重要」

選擇其中一個作為題目（必須抄題），根據你的自身經驗或見聞，闡述你對知識與想像力的看法。（占 50 分）

試題結束

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英文試題

作答提示：

請根據題組內容，在答案卷上用英文寫出正確、完整的答案（英翻中除外）

題組一：18 分

Imagine having a watch that could allow you to listen to the sound or watch the video of your heart as it beats in your chest. Researchers have taken the first step down that road. They have just invented a wearable ultrasound patch, a stick-on sonar, which offers a flexible way to see deep inside the body.

Ultrasound maps tissues and fluids by recording how sound waves bounce off of them. That tool can help doctors find signs of organ damage or cancer quickly. It can even track bacteria. Unfortunately, most ultrasound machines are heavy, bulky, and not **portable**. There are wearable ones, but usually they can't spot details or can be used for only short periods.

The new patch can work for up to 48 hours straight—even while you are moving around a lot, like when you play a sport. Plus, this patch sees just as well as the machines in hospitals.

“This is just the beginning,” says Xuanhe Zhao, a mechanical engineer at the Massachusetts Institute of Technology (MIT) in Cambridge. He was part of the team that created the new device. Next, Zhao's team will make the patch wireless so it can talk to phones.

“This has a lot of medical possibilities. Say you stick a patch over someone's heart. **The images it would take might help predict heart attacks months before the disaster hits,**” says Aparna Singh, a biomedical engineer at Columbia University in New York City who did not take part in this research. Placed on a COVID-19 patient, the patch—which is only about the size of a quarter coin—might become an easy way to find lung problems as they develop.

In many poorer countries, going to hospitals is really hard, and there aren't any high-power imaging machines to help the patients there. But the new device “has a huge potential” to help people in these parts of the world. Right now, the patch costs about \$100 to make. But its designers hope to bring that cost down.

 bounce 彈跳	bulky 巨大笨重的	device 裝置	flexible 有彈性的
fluid 體液	mechanical 機械的	potential 潛力	sonar 聲納
ultrasound 超音波			

1. (T / F) The wearable ultrasound patch may help watch the changes of COVID-19 patients' lungs.
2. It can be inferred from the second paragraph that when something is **portable**, it _____.
 - (A) is common and can be found everywhere
 - (B) is very useful and guarantees satisfaction
 - (C) is so easy and simple that everyone can use it
 - (D) can be carried or moved around easily
 - (E) can be put on quickly by people of all ages
3. What can we learn about the wearable ultrasound patch from the passage?
 - (A) It can find not only bacteria but also viruses.
 - (B) It can be used when you exercise in a gym.
 - (C) It is developed only by a group of doctors.
 - (D) Its designers think its price should be higher.
 - (E) Its only advantage is its size and convenience.
4. What object does the author use to help readers understand how big a wearable ultrasound patch is?
5. According to the passage, the patch will be able to talk to phones after Zhao's team makes it _____.
6. Please translate the following sentence into Chinese: **“The images it would take might help predict heart attacks months before the disaster hits.”**

題組二：16 分

Antioxidants are chemicals that are widely used in consumer products and some are connected with harmful effects. People can come into contact with antioxidants through skin, nose, or mouth. Antioxidants are added to a variety of consumer products, such as food packaging and personal care products, because they help to keep packaging materials stable. While antioxidants can be found in consumer products, they are not chemically bonded to the materials they are added to, leading to their migration into the environment. In addition to consumer products, antioxidants have been found in environmental samples such as air particles, waste dust, and sediment.

Common sources of antioxidants include:

- Personal care products: antioxidants such as butylated hydroxyanisole (BHA) and butylated hydroxytoluene (BHT) can be used as a preservative in lip products, hair products, sunscreens, and perfumes.
- Food packaging: BHT and Irganox are both types of antioxidants that are added to plastics like polyethylene and polypropylene. Antioxidants in plastics have been found to migrate into oil and food that the plastic comes into contact with. BHT can also be added to wax lining of food packaging and has been found to migrate into food.
- Food: while many foods contain natural antioxidants, which can provide health benefits, synthetic antioxidants such as BHA and BHT are often added to foods to prevent fats and oils from going bad.

While studies on their harmful effects are limited, their findings have been worrying. In laboratory studies, BHT caused liver cell damage. BHA caused cancer in laboratory animals and is considered a possible carcinogen. Antioxidants can also turn into other harmful chemicals. When BHT comes in contact with oxygen, it can turn into a chemical called BHT-Q. BHT-Q is linked with visual conditions in humans and neurotoxic effects in animals.

We can reduce exposures of antioxidants for everyone by cutting the use of plastic and replacing harmful antioxidants with safer substitutes. We can let our favorite retailers, brands, and policymakers know that we want them to ban harmful antioxidants and make sure substitutes are safer. On the other hand, we should avoid products that list antioxidants as ingredients and eating fewer processed foods may also reduce exposure to antioxidants used in such foods.

	ban 禁止	bond 結合	consumer 消費者	exposure 接觸
ingredient 成份	neurotoxic 毒害神經的	particle 微粒	preservative 防腐劑	
retailer 零售商	sediment 沉澱	substitute 代替品	sunscreen 防曬乳	
synthetic 合成的				

1. (T / F) Some antioxidants are beneficial to human beings, like those natural antioxidants that can be found in personal care products.
2. What can we learn from the passage about antioxidants?
 - (A) Irganox is a type of natural antioxidants that is often added to plastics like polyethylene and polypropylene.
 - (B) Antioxidants are hard to get rid of because they are firmly fixed with the materials they are added to.
 - (C) Human beings usually get exposed to antioxidants mainly through touching, breathing, and eating.
 - (D) Antioxidants are not usually found in lip balms and hair conditioners because they don't go bad easily.
 - (E) Synthetic antioxidants are added to foods to boost their health benefits and stop fats and oils from spoiling.
3. Which of the following aspects about antioxidants is **NOT** discussed in the passage?
 - (A) Their functions.
 - (B) Their sources.
 - (C) Their effects.
 - (D) Their history.
 - (E) Their properties.
4. According to the passage, what health issue is connected with the harmful chemical that is produced when BHT is touched upon by oxygen?
5. Choose a word from the passage that best completes the following sentence and make changes if necessary: For lack of chemical bonds, antioxidants that are added to packaging materials tend to _____ into the environment.
6. According to the passage, what are the two things that the author advises readers to do with their diet to avoid antioxidants in foods?

題組三：16 分

The continents have been changing and moving over millions of years. This movement is caused by plates under the Earth's surface, which are like big puzzle pieces that fit together to make up the Earth. Think of it like this: if you took a big toy map of the world and pushed different pieces together or pulled them apart, you would see how the continents can change over time. This is how the Earth has changed and continues to change over millions of years, shaping our planet and creating the world we live in today!

Asia, Europe, and America have been affected by this process in different ways. Over the last 200 million years, Asia and Europe have been part of the same landmass, called Pangea, before splitting apart and moving to their present-day locations. America was once part of Pangea as well, but began to separate from it around 165 million years ago and move toward its current position. This movement of the continents continues today and is responsible for the formation of mountain ranges, earthquakes, and volcanic activity.

There is evidence of this movement in many different forms. For example, matching rock formations and fossils can be found on different continents, showing that they were once connected. **The continents also fit together like puzzle pieces, which is further evidence of their past movement.**

The formation and movement of the continents over time has had a huge effect on the distribution of different human populations and has brought about the development of different races and cultures. As the continents separated and shifted, human populations became isolated from each other. Over time, these isolated populations developed unique adaptations to their environment, such as skin color and other physical traits. As a result, different human populations developed different racial characteristics, leading to the concept of different races.

However, it is important to understand that the separation of continents did not determine the development of races, but rather the evolution of human populations that already settled on the continents. The idea of different races being separate and different from each other is a cultural idea and not supported by scientific evidence.

	adaptation	適應	characteristic	特色	concept	概念	distribution	分佈
	evolution	演化	formation	形成	fossil	化石	isolated	被分離的
	shift	移動	split	分裂	trait	特徵	volcanic	火山的

1. (T / F) Volcanic activities cause the plates under the surface to move and in turn the continents to shift.
2. In which paragraph does the sentence “The sandstone formations in the desert region of the southwestern United States are similar to those found in the deserts of southern Africa.” most likely appear?
 - (A) Paragraph 1.
 - (B) Paragraph 2.
 - (C) Paragraph 3.
 - (D) Paragraph 4.
 - (E) Paragraph 5.
3. According the passage, which of the following can serve as direct evidence for continental shifts?
 - (A) Similar rates of population growth.
 - (B) Matching fossils on different lands.
 - (C) Puzzles found on different continents.
 - (D) Genetic differences of different races.
 - (E) Similar cultures in different places.
4. What was the name of the landmass that Asia and Europe were part of?
5. What causes the formation of mountain ranges, earthquakes and volcanic activity?
6. Please translate the following sentence into Chinese: **“The continents also fit together like puzzle pieces, which is further evidence of their past movement.”**

試題結束
